

SMARTEXAM PORTAL

Comprehensive Assessment Question Bank

Prepared for Student Evaluations

This bank contains curated questions covering:

- Campus Recruitment Training (CRT) - 25 Questions
- Java Basics MCQ - 20 Questions
- HTML & CSS MCQ - 25 Questions
- Java Basic Coding Tasks - 4 Questions (each with 2 Test Cases)

Designed to be importable and fully compatible with the SmartExam Platform.

Generated on: August 2026

PART A: Campus Recruitment Training (CRT) - 25 MCQs

Q1. A train 120m long passes a telegraph post in 6 seconds. Find the speed of the train in km/h.

(a) 36 km

Q2. A can complete a piece of work in 12 days and B can complete the same work in 24 days. How many days will they take if they work together?

(a) 6 days

Q3. What is the percentage change in the area of a rectangle if its length is increased by 20% and its width is decreased by 10%?

(a) 8% inc

Q4. A sum of money triples itself in 10 years at simple interest. What is the rate of interest per annum?

(a) 10%

Q5. A bag contains 5 red and 7 blue balls. If a ball is drawn at random, what is the probability that it is red?

(a) 5/12

Q6. Find the average of the first 50 natural numbers.

(a) 25 (l

Q7. The ratio of ages of A and B is 3:4. If the sum of their ages is 28 years, find B's age.

(a) 12 ye

Q8. An article is sold for \$180 at a loss of 10%. What is its cost price?

(a) \$200

Q9. Find the Highest Common Factor (HCF) of 48 and 72.

(a) 12 (l

Q10. In how many different ways can the letters of the word 'CAT' be arranged?

(a) 3 (b)

Q11. Find the missing number in the sequence: 2, 5, 10, 17, 26, ?

(a) 35 (l

Q12. If RED is coded as 360, how is BLUE coded?

(a) 480

Q13. Pointing to a photograph, a man said, 'I have no brother or sister but that man's father is my father's son.' Who is in the photograph?

(a) His son

Q14. Find the odd one out among the following shapes:

(a) Square

Q15. Find the missing letters in the pattern: AZ, BY, CX, ?

(a) DW

Q16. A man walks 5 km East, turns right and walks 4 km, then turns left and walks 5 km. In which direction is he from his starting point?

(a) North-

Q17. Identify the logical conclusion: 'All men are mortal. Socrates is a man.'

(a) Socrate

Q18. If '+' means 'x', '-' means '+', 'x' means '/' and '/' means '-', then what is: $10 + 5 - 3 / 2$?

(a) 51 (

Q19. Which number replaces the question mark in the sequence? 1, 8, 27, 64, ?

(a) 125

Q20. In a row of 20 students, A is 8th from the left. What is A's position from the right?

(a) 12th

Q21. Choose the synonym of 'Candid'.

(a) Frank

Q22. Choose the antonym of 'Giant'.

(a) Dwarf

Q23. Identify the grammatical error in this sentence: 'Each of the students have submitted their assignment.'

(a) Each o

Q24. Complete the sentence: 'She is _____ of dogs.'

(a) afraid

Q25. What is the meaning of the idiom 'Burn the midnight oil'?

(a) Work

PART B: Java Basics - 20 MCQs

Q1. Which component of Java is responsible for running the compiled bytecode?

(a) JDK

Q2. Which data type is used to store fractional numbers in Java by default?

(a) float

Q3. Which keyword is used to prevent method overriding in Java?

(a) static

Q4. What is the size of the int data type in Java?

(a) 8-bit

Q5. Which of the following is NOT an Object-Oriented Programming (OOP) concept in Java?

(a) Polym

Q6. Which memory area stores local variables in Java?

(a) Heap

Q7. What is the return type of the hashCode() method in the Object class?

(a) String

Q8. Which package is automatically imported by default in every Java program?

(a) java.io

Q9. What happens when a thread's sleep() method is called?

(a) Threa
indefinite

Q10. Which exception is thrown when an array is accessed with an invalid index?

(a) NullP
IllegalArg

Q11. Which keyword is used to inherit a class in Java?

(a) imple

Q12. Can we overload the main method in Java?

(a) Yes

Q13. Which interface from the Java Collections Framework does NOT allow duplicate elements?

(a) List

Q14. Which class is used to create mutable strings in Java?

(a) String

Q15. What is the default value of a boolean instance variable in Java?

(a) true

Q16. Which operator is used to check if an object is of a specific class type?

(a) instan

Q17. What is a default constructor of a class?

(a) Creat
paramete

Q18. Which statement is true about interface variables in Java?

(a) They a
by implem

Q19. Which block is guaranteed to execute whether an exception is caught or not?

(a) try (f

Q20. What is the purpose of garbage collection in Java?

(a) Deleti
stack fran

PART C: HTML & CSS - 25 MCQs

Q1. What does HTML stand for?

(a) Hyper
High Text

Q2. Which HTML element is used for the largest heading?

(a) <h6>

Q3. Which tag is used to create a hyperlink in HTML?

(a) <link>

Q4. What is the correct HTML element for inserting a line break?

(a) <lb>

Q5. Which attribute is used to provide alternative text for an image if it cannot be displayed?

(a) title

Q6. Which element is used to define an unordered list in HTML?

(a)

Q7. In CSS, what does 'CSS' stand for?

(a) Creati

Q8. Which CSS property is used to change the background color of an element?

(a) color

Q9. How do you select an element with the ID 'header' in CSS?

(a) .head

Q10. Which CSS property controls the text size?

(a) font-st

Q11. What is the default value of the position property in CSS?

(a) absolu

Q12. Which CSS property is used to create space inside an element's border?

(a) margin

Q13. How do you center a block-level element horizontally using margins?

(a) margin

Q14. Which CSS flexbox property aligns items vertically when flex-direction is row?

(a) justify

Q15. What is the correct HTML for adding a background color to a web page?

(a) <body
<body col

Q16. Which HTML tag is used to define an internal style sheet?

(a) <css>

Q17. Which CSS property is used to make text bold?

(a) font-st

Q18. How do you display hyperlinks without an underline in CSS?

(a) a {text

Q19. Which HTML5 element is used to represent self-contained content, like a blog post or news article?

(a) <secti

Q20. What is the correct CSS syntax to make all <p> elements uppercase?

(a) p {tex
uppercas

Q21. Which property specifies the stack order of an element (which element should be placed in front of or behind others)?

(a) x-inde

Q22. Which HTML element is used to group inline-elements?

(a) <div>

Q23. What does the CSS Grid property 'grid-template-columns: repeat(3, 1fr);' do?

(a) Create

Q24. Which of the following CSS selectors has the highest specificity?

(a) Class

Q25. How do you write a comment in CSS?

(a) // this

PART D: Java Coding Tasks - 4 Questions (2 Test Cases)

Coding Problem 1: Reverse a String

Write a Java method that takes a string as input and returns the reversed string.

Method Signature: `public String reverse(String str)`

Code Template:

```
public class Solution {  
    public String reverse(String str) {  
        // Write your code here  
    }  
}
```

Test Cases:

Test Case 1: Input: "hello" => Expected Output: "olleh"

Test Case 2: Input: "Java" => Expected Output: "avaJ"

Coding Problem 2: Check Prime Number

Write a Java method that returns true if a given integer n is prime, and false otherwise. Assume $n > 1$.

Method Signature: `public boolean isPrime(int n)`

Code Template:

```
public class Solution {  
    public boolean isPrime(int n) {  
        // Write your code here  
    }  
}
```

Test Cases:

Test Case 1: Input: 7 => Expected Output: true

Test Case 2: Input: 10 => Expected Output: false

Coding Problem 3: Find Maximum in Array

Write a Java method that finds and returns the largest element in an integer array.

Method Signature: `public int findMax(int[] arr)`

Code Template:

```
public class Solution {  
    public int findMax(int[] arr) {  
        // Write your code here  
    }  
}
```

Test Cases:

Test Case 1: Input: `new int[]{1, 5, 3, 9, 2}` => Expected Output: 9

Test Case 2: Input: `new int[]{-3, -1, -7, -2}` => Expected Output: -1

Coding Problem 4: Factorial of a Number

Write a Java method that returns the factorial of a given non-negative integer n.

Method Signature: public int factorial(int n)

Code Template:

```
public class Solution {  
    public int factorial(int n) {  
        // Write your code here  
    }  
}
```

Test Cases:

Test Case 1: Input: 5 => Expected Output: 120

Test Case 2: Input: 0 => Expected Output: 1